

DepBoxia: Depth Perception Training in Boxing, an Immersive Approach

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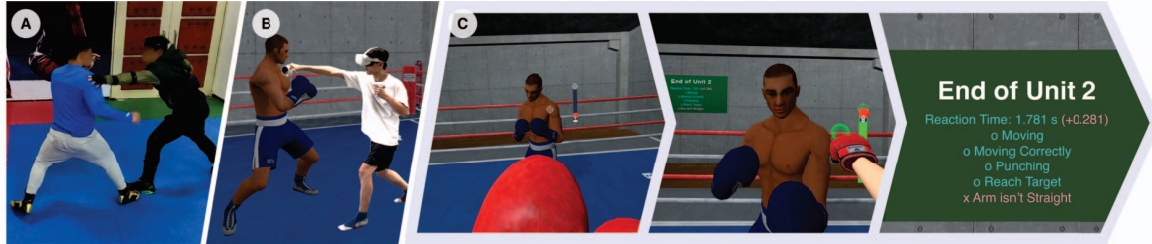


Figure 1: (A) Touch shoulders exercise, a traditional exercise to train players' distance management ability, also known as depth perception. (B) The immersive depth perception training system allows players to train in virtual reality on their own, using only one head-mounted device and two handheld controllers. (C) The immersive training system that integrates visual and audio guidance, along with sophisticated feedback mechanisms to enhance the overall training experience.

ABSTRACT

Depth perception, the ability to interpret an opponent's punches in terms of timing and distance, is fundamental for novice boxer training. Traditional methods rely on coach support and teamwork, leading to a high entry barrier for novice boxers to train whenever or wherever. To address this, we propose an immersive training system using virtual reality (VR) that integrates visual and audio guidance, specifically designed to enhance depth perception in boxing for novice boxers. Our rigorous experiment shows significant improvements in accuracy and reaction time. We introduce the system to professional boxing coaches, enabling them to integrate a systematic approach to training novice boxers' depth perception.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI;

1 INTRODUCTION

In order to win a boxing match, a boxer must analyze their opponent's punches, considering timing, distance, angle, and speed, using depth perception. This skill enables accurate spatial assessment, guiding informed decisions and precise actions in the match. Traditional training method is constrained by the need for cooperation with a coach and teammate, limited to space and time. In presence training methods, novice boxing players can only attempt to comprehend the concept of depth perception solely through verbal instructions from the coach, which can be challenging.

While previous studies have explored the application of virtual reality in sport training [2, 3, 5], some have even extended their endeavors into the realm of boxing [1, 6]. However, none have specifically focused on depth perception training. To overcome

the constraints of traditional training methods, we introduce an immersive training approach using virtual reality tailored to depth perception in boxing. This approach combines visual and audio feedback to enable a comprehensive understanding of depth during training. Utilizing virtual reality, we prevent training injuries for novice boxers. Our immersive feedback system improves distance perception and skills. It's user-friendly and allows independent training with no prior expertise needed.

2 IMMERSIVE DEPTH PERCEPTION TRAINING SYSTEM

The immersive depth perception training system was developed on the Unity3D platform using the Oculus Integration SDK and C#. We use FinalIK [4] to minimize sensor use, relying on only two handheld controllers and one head-mounted device for straight-arm evaluation.

2.1 Formative Study

We conducted an exploratory study with a former national boxing team coach to explore the knowledge of boxing and its training methods. The coach emphasized that using a straight hand to target the opponent's scoring areas, including the shoulder and waist, was crucial for scoring valid points. To achieve this, the player needs a strong sense of depth perception to accurately gauge the distance from the opponent and their arm's length. The coach also showed how they train boxers in depth perception with the "touch shoulders" exercise (as shown in Fig. 1(A)), which involves maintaining an ideal distance from the opponent, neither too close nor too far. The exercise consists of three 1.5-minute rounds per day, with players facing different opponents using varying stances. Following the coach's professional advice, novice boxers can significantly improve depth perception and react appropriately to their opponents' spatial positions by completing around 15 practice rounds against opponents with different stances.

2.2 System User Flow

The system's user flow is outlined in Fig. 2: In the initial step, calibration ensures safety by setting the player's real-life movable range. Use controllers to measure the user's height and arm length. After calibration is complete, the system enters training with seven units per turn. Each unit starts with a 4-second ready step. Then, in

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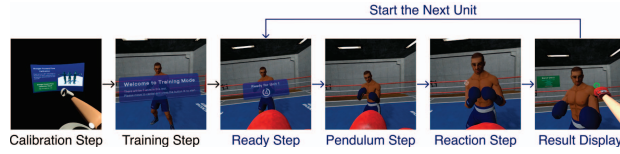


Figure 2: User flow of the depth perception training system.

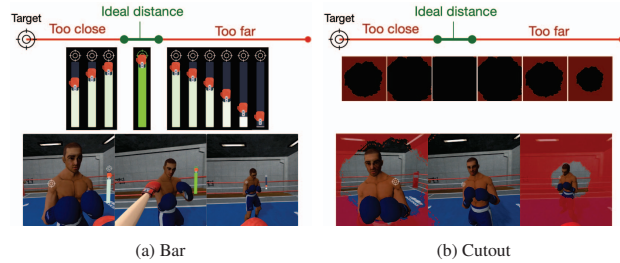


Figure 3: Distance guidance variation.

the pendulum step, the virtual opponent moves back and forth. After 1 to 2 seconds, the virtual opponent advances or retreats, displaying a target icon. The player adjusts their position for a straight-arm hit. Successful hits complete the reaction step, with results displayed. The process repeats for each unit.

2.3 System Mechanisms Design

To replicate lifelike reactions similar to real-world training and emulate the presence of a virtual coach offering performance recommendations, we integrated multiple feedback mechanisms to enhance the authenticity of interactions within the virtual reality environment. These mechanisms include instant visual and audio responses when the player hits the virtual opponent, as well as performance suggestions presented through visual displays and voice comments to guide the player's improvement as shown in Fig. 1(C).

Additionally, we leveraged the advantages of VR to provide extra cues and guidance, aiding players in improving their performance and understanding of depth perception in boxing. This includes the integration of two dynamic visual aids for precise distance perception guidance: bar (represented by an energy bar) and cutout (indicated by a varying cutout area), which adjust based on the player's distance from the virtual opponent, signifying the ideal punching moment and distance, as shown in Fig. 3. Alongside the visual distance guidance, we provide an auditory cue, a bell sound, 0.284 seconds ahead of the assigned standard reaction time to prompt users to react faster (for more design details, please refer to the supplementary).

3 EXPERIMENT

3.1 Experiment Design

We used Meta Quest 2 for indoor depth perception training, emphasizing high accuracy and speed, evaluating with accuracy and reaction time metrics. The 32 participants (16 males, 16 females) lacked professional boxing training, had normal vision, and averaged 24.45 years ($SD = 3.56$). Self-reported VR familiarity varied: 46.9% rated below 3, 40.7% above 5 on a 7-point Likert scale (1=not familiar, 4=neutral, 7=very familiar), mitigating potential VR familiarity bias.

The experiment spanned three weeks, comprising three sessions: a pre-test session in which each participant completed two consecutive trials with no additional guidance; a training session where each participant performed seven units per trial, four trials a day for four days with the extra guidance; and a post-test session in which

Table 1: Paired-samples t-tests on accuracy and reaction time.

		Accuracy (%)		Reaction Time (s)	
	N	Mean	StDev	Mean	StDev
Pre-Test	32	38.62	19.86	1.428	0.399
Post-Test	32	66.52	23.27	1.273	0.268
Difference	32	27.90	24.37	-0.156	0.327
<i>(t-value, p-value)</i>		<i>(-6.478, 0.000*)</i>		<i>(2.693, 0.011*)</i>	

participants repeated the same two trials as in the pre-test session. Additionally, during the final week, we administered a questionnaire survey to collect participants' subjective experiences regarding the system.

3.2 Hypotheses

Here are the hypothesis of our work:

1. The proposed immersive training system will significantly enhance novice boxing players' depth perception.
2. The proposed system gamifies and enhances training enjoyment, boosting engagement and motivation for novice boxers.

3.3 Experiment Result

We performed paired-samples t-tests on accuracy and reaction time pre-test vs. post-test data at a 95% confidence level. From the statistical results shown in Table 1, both p-values are below 0.05, signifying significant metric improvements.

After using the system, 91% of participants were willing to use it for boxing training. They found it beginner-friendly, convenient, and flexible in terms of time and space. The system's customizable training options, resembling a video game, made training interesting and enjoyable, boosting motivation. Additionally, most participants showed an increased eagerness to learn more about boxing, highlighting the system's success in stimulating interest and engagement (for more subjective opinions on the system, please refer to the supplementary).

4 CONCLUSION

Our research pioneered a tailored VR system for unexplored depth perception training in boxing, enhancing performance. Validating its benefits enables integration into training programs, offering athletes cutting-edge technology.

REFERENCES

- [1] H. Chen, Y. Wang, and W. Liang. Vcoach: Enabling personalized boxing training in virtual reality. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 546–547, 2022. doi: 10.1109/VRW55335.2022.00124
- [2] C. Ku, J.-J. Weng, Y.-H. Wang, D.-X. Wu, Y.-M. Lau, W.-L. Tsai, T.-C. Wu, H.-K. Chu, and M.-C. Hu. Table tennis skill learning in vr with step by step guides using forehand drive as a case study. In *2022 IEEE International Conference on Artificial Intelligence and Virtual Reality (AIVR)*, pp. 275–282. IEEE, 2022.
- [3] T. Lin, Y. Yang, J. Beyer, and H. Pfister. Sportsxr-immersive analytics in sports. *arXiv preprint arXiv:2004.08010*, 2020.
- [4] RootMotion, Inc. *RootMotion - Final IK*, 2016.
- [5] W.-L. Tsai, L.-w. Su, T.-Y. Ko, C.-T. Yang, and M.-C. Hu. Improve the decision-making skill of basketball players by an action-aware vr training system. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 1193–1194. IEEE, 2019.
- [6] W. Xu, H.-N. Liang, X. Ma, and X. Li. Virusboxing: A hiit-based vr boxing game. In *Extended Abstracts of the 2020 Annual Symposium on Computer-Human Interaction in Play, CHI PLAY '20*, p. 98–102. Association for Computing Machinery, New York, NY, USA, 2020. doi: 10.1145/3383668.3419958